

Aman Soni

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EDUCATION

University of Michigan

Master of Science in Computer Science and Engineering; CGPA: 3.74/4

Ann Arbor, MI

Aug 2025 – Dec 2026

Relevant Coursework: NLP, Data Structures & Algorithms, Scalable and Distributed Systems, AI Foundations

SRM Institute of Science and Technology, Kattankulathur

Chennai, India

Bachelor of Technology in Computer Science and Engineering; CGPA: 9.51/10

June 2018 – May 2022

WORK EXPERIENCE

Delta Dental of Michigan

East Lansing, MI

Software Engineer Intern

May 2026 – Aug 2026

- Built Claims History Visualizer, an integrated dental-claims intelligence platform using FastAPI, Pandas, Plotly, SQLAlchemy, and PostgreSQL pipelines to unify patient data, render tooth-level treatment timelines, and enable fast search by procedure, date, and anatomy - improving reviewer lookup speed by ~50% across 100K+ records.
- Reduced manual case-review time by ~30% by streamlining claims data pipelines integrating PostgreSQL and Excel sources into a unified review interface.
- Added AI-assisted triage using a PyCaret AutoML workflow that benchmarks classification models on historical claims patterns to flag high-risk manual-review cases, improving adjudication consistency by ~20%.

Aditya Birla Capital Ltd.

Mumbai, India

Software Engineer

July 2023 – July 2025

- Led development and management of personal loan web applications, streamlining processing of 700 loans daily with USD 3,000 typical loan amount.
- Engineered REST APIs using Python to integrate modules like Bureau Enquiries, Fraud Checks, KYC Verification, E-Mandate, and Disbursal, automating checks and reducing operational turnaround time by 25%.
- Built a GraphQL-based MIS portal for generating customer onboarding reports, enabling 15+ business stakeholders to access real-time data and optimize the marketing funnel - contributing to a 10% increase in successful loan applications.

Graduate Engineer Trainee

July 2022 – June 2023

- Engineered high-throughput asynchronous processing modules using RabbitMQ, ensuring 100% reliability for 60,000+ monthly financial document deliveries.
- Executed a streamlined data pipeline to migrate 10+ on-premise database tables to AWS RDS, leveraging stored procedures, SFTP, and Lambda triggers to automate daily transfers and reduce manual effort.

PROJECTS

TriGuard: Validity-Aware Semantic Caching for LLM APIs

Jan 2026 – May 2026

- Built TriGuard, a 3-gate semantic caching middleware for LLM APIs using Redis + ChromaDB dual storage, reducing false positive cache hits by 30% and cutting operational costs 14x over no-cache deployment.
- Trained an adaptive TTL classifier (Logistic Regression + BGE embeddings) achieving 97% recall on volatile query detection, preventing stale data propagation for time-sensitive queries.
- Replaced cosine similarity with Pythagorean Fuzzy Sets (PyFS) and deployed asynchronous Triple SLM verification (Qwen2.5-1.5B) to filter hallucinations pre-caching, sustaining 44% hit rate at ~75ms warm-path latency on A100 GPUs.

MUSE (Machine enhanced User Sound Experience)

Aug 2025 – Dec 2025

- Developed "MUSE" - an AI system leveraging Demucs, a deep learning model for music source separation (vocals, drums, bass, instruments) achieving 90%+ SDR using Python, PyTorch, Librosa.
- Built an end-to-end interactive platform integrating Flask, React, and PostgreSQL to allow users to upload audio, visualize separated tracks, and receive personalized song recommendations generated through content-based and user-feedback algorithms, improving recommendation relevance by 30% in testing.

SKILLS

- Languages:** C++, C, Python, SQL, JavaScript, TypeScript, Java, Go
- Frameworks:** Flask, Falcon, Django, FastAPI, Node.js, React, Spring Boot, Scikit-learn, PyTest, Pandas, NumPy, LangChain
- Cloud & Databases:** AWS Services (DynamoDB, RDS, S3, Lambda, EC2), MySQL, PostgreSQL, MongoDB, Azure
- Platforms & Tools:** Linux, Docker, GIT, Bitbucket, JIRA, Jenkins, Kafka, Redis, RabbitMQ, GitHub CoPilot, Claude, Cursor
- Domains:** Full-Stack Web Development, Machine Learning, Object-Oriented Design, CI/CD, Testing Practices, Microservices, Agile Environments, Code Review, Software Development Lifecycle, Agentic AI